



SCHOOL OF COMPUTER TECHNOLOGY

AASD 4008 Big Data Tools and Technologies II

Introduction to Alteryx



Definition: Alteryx is a data analytics platform that streamlines data blending, preparation, and analytics processes.

Purpose: It empowers users to handle complex data tasks efficiently, facilitating data-driven decision-making.

Use Cases: Alteryx is used for ETL, data cleansing, predictive analytics, spatial analysis, and more.

What is Alteryx ?



Definition: Alteryx is an end-to-end analytics platform for data blending, preparation, and advanced analytics.

Key Features:

- Data cleansing and transformation
- Spatial and predictive analytics
- Workflow automation

Alteryx is a comprehensive end-to-end analytics platform designed to **empower users** with **advanced capabilities** in data blending, preparation, and analytics. It provides a user-friendly interface with a visual workflow that simplifies complex data tasks, making it accessible to both data **professionals** and **business users**. Key features include data cleansing, spatial and predictive analytics, and workflow automation.

Alteryx Workflow - Pipeline

Workflow Concept:

- Visual representation of data processes.
- Drag-and-drop tools for data manipulation.

Connectivity:

- Easily connect data sources and tools.

The Alteryx **workflow(pipeline)** is a visual representation of data processes, allowing users to drag-and-drop tools onto the **canvas** for seamless data manipulation. The platform promotes connectivity by facilitating easy linking of data sources and tools, creating a streamlined and efficient data workflow



Alteryx - ETL



Alteryx is a robust ETL (Extract, Transform, Load) tool that revolutionizes data integration processes. Offering a visually intuitive interface, Alteryx empowers users to seamlessly connect to diverse data sources, transform and cleanse data with a rich set of tools, and efficiently load processed data into target systems or data warehouses. With features like in-database processing, automation, and scheduling, Alteryx accelerates ETL workflows, allowing organizations to enhance data quality, streamline migration projects, and facilitate the efficient management of data for analytics and reporting purposes. Its user-friendly design ensures accessibility for both technical and non-technical users, making Alteryx a versatile solution for organizations seeking to optimize their data integration processes.

Alteryx – Bigdata Analytics



Alteryx stands as a pivotal tool in the realm of **big data analytics**, seamlessly integrating with robust platforms like **Hadoop** and **Spark**. Its advanced capabilities empower organizations to navigate the complexities of vast and diverse datasets efficiently. Alteryx facilitates data blending, predictive analytics, and spatial analysis on large-scale data, providing users with the tools needed to extract actionable insights from the ever-expanding world of big data. With **scalability** at its core, Alteryx ensures that businesses can process and derive value from **massive datasets**, positioning itself as a key player in the landscape of modern big data analytics.

Core Alteryx Tools



- **Input Tools:** Connect to data sources

Alteryx offers a variety of input tools to connect to various data sources, including databases, spreadsheets, and cloud services.

- **Transform Tools:** Clean, filter, and manipulate data.

These tools empower users to clean, filter, and manipulate data, providing a wide range of options for data transformation.

- **Output Tools:** Save or export data.

Alteryx supports diverse output options, enabling users to save or export processed data to different formats or databases.

Spatial and Predictive Analytics



- **Spatial Tools:** Geospatial analysis for mapping.

Alteryx includes a robust set of spatial tools for geospatial analysis, allowing users to incorporate location-based insights into their analytics workflows.

- **Predictive Tools:** Machine learning for forecasting and modeling.

The platform's predictive tools facilitate machine learning and advanced analytics, enabling users to build models, make predictions, and derive valuable insights from their data.

Use Case – Marketing Campaign



- **Scenario:** Analyzing customer data for targeted marketing.
Imagine a scenario where a marketing team wants to analyze customer data for a targeted marketing campaign.
- **Alteryx Usage:** Data cleaning, segmentation, predictive modeling.
Alteryx can be utilized for data cleaning, segmentation, and predictive modeling to enhance the effectiveness of the marketing campaign.
- **Outcome:** Improved campaign efficiency and ROI.
The outcome includes improved campaign efficiency, higher conversion rates, and a better return on investment (ROI).

Use Case – Financial Analysis



- **Scenario:** Analyzing financial data for insights.

Consider a financial analysis scenario where comprehensive insights are needed to make informed financial decisions.

- **Alteryx Usage:** Data blending, time series analysis.

Alteryx facilitates data blending, time series analysis, and other relevant techniques to derive meaningful insights for enhanced financial decision-making.

- **Outcome:** Enhanced financial decision-making.

The outcome involves more accurate financial forecasting and better strategic financial planning.

Pros of Alteryx



- **User-Friendly:**Intuitive drag-and-drop interface.

Alteryx's intuitive drag-and-drop interface makes it accessible to users with varying technical expertise.

- **Versatility:**Supports a wide range of data tasks.

The platform supports a wide range of data tasks, from basic data cleaning to advanced predictive analytics.

- **Community Support:**Active user community and resources.

Alteryx boasts an active and supportive user community, providing valuable resources, forums, and knowledge sharing.

Cons of Alteryx



- **Learning Curve:** Advanced features may have a learning curve. While Alteryx is user-friendly, certain advanced features may have a learning curve for users unfamiliar with data analytics tools.
- **Cost:** Licensing costs may be a consideration for smaller budgets., particularly for smaller budgets or individual users (desktop license cost is around 8k per year)

Competitors



•**Tableau Prep:** Focuses on data preparation with a user-friendly interface tailored for Tableau users.

Knime: An open-source analytics platform with a modular design, allowing users to create and execute data workflows.

Trifacta: cloud-based data wrangling tool designed for exploring, cleaning, and enriching diverse datasets.



Demo !



Creating a workflow (pipeline) with CSV data with Alteryx